

## MANAS JOSHI

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Honors Computing Science student focused on real-time ML systems and backend engineering. Experienced building low-latency inference pipelines, REST APIs, and hardware-integrated ML applications. Seeking SWE or AI-focused co-op roles.

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## TECHNICAL SKILLS

**Languages:** Python, C++, Java, C

**Backend & Systems:** Flask, Django, REST APIs, Linux, Asynchronous Programming

**AI/ML:** PyTorch, TensorFlow, Scikit-learn

**Databases & Tools:** MySQL, Git, Docker, UNIX

**Core CS:** Data Structures, Algorithms, OOP, Complexity Analysis

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## PROJECTS

**NeuroMotion** — Python, PyTorch, Flask, JavaScript, HTML, CSS — **Nov 2025**

- Designed and deployed a real-time EEG classification pipeline with sub-200ms latency.
- Built and deployed a RESTful Flask backend serving scalable real-time model inference.
- Implemented sliding-window signal processing with asynchronous prediction handling.
- Integrated OpenBCI hardware into a scalable data ingestion pipeline.
- Achieved 87% validation accuracy; 2nd place at NatHacks.

**SkyNav** — Python, Scikit-learn, Flask, JavaScript, HTML, CSS — **Feb 2025**

- Implemented A\* pathfinding with dynamic weather constraints using priority queues and heuristic optimization.
- Designed a modular backend supporting real-time route recomputation.
- Integrated OpenWeather API with asynchronous request handling.
- Built REST endpoints powering live UI updates.

**Math-Vision-Model** — Python, TensorFlow, Matplotlib — **Jul 2024 – Jan 2025**

- Engineered a ResNet-based image classification pipeline achieving 88% accuracy across 89 classes.
- Built a reproducible training workflow with versioned model exports (.h5).
- Automated dataset preprocessing combining EMNIST and synthetic data generation.
- Improved model robustness through hyperparameter tuning and structured evaluation.

**natHACKS 2023 – Video Game** — Unity, C#, C++ — **Nov 2023**

- Designed reinforcement learning-based control strategies for EMG-driven prosthetic assistance.
  - Implemented signal preprocessing and real-time EMG feature extraction pipelines.
  - Evaluated policy performance using the RL-Glue experimentation framework.
  - Collaborated across EE, CS, and biomedical teams on assistive robotics integration.
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## EXPERIENCE

**Bionix · EMG & RL Systems** — Volunteer — **Oct 2025**

Edmonton, AB

- Developed RL-based control logic for EMG-driven prosthetic leg assistance.
  - Collected and labeled EMG datasets; evaluated control policies using RL-Glue.
  - Collaborated in a multidisciplinary team spanning EE, CS, and biomedical domains.
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## EDUCATION

**University of Alberta** — BSc (Honors) Computing Science — **Expected May 2027**

Relevant Coursework: Machine Learning, Reinforcement Learning, Algorithms, Computer Architecture